NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

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Course: **Artificial Intelligence**	ASSIGNMENT- 2	**DUE DATE: 29/03/2026**
Course Code: **AI2002**		Total Questions: 7 **Total Marks: 100**

Instructions:

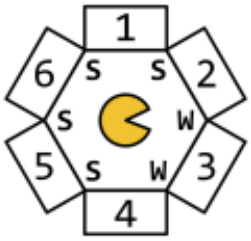
1. Attempt **all questions by hand**.
2. **Scan your handwritten work** and submit as a **PDF** on GCS.
3. No plagiarism—copied work gets zero.

Question 1: [15 Marks]

Q1. Pacman is trapped! He is surrounded by mysterious corridors, each of which leads to either a pit (P), a ghost (G), or an exit (E). In order to escape, he needs to figure out which corridors, if any, lead to an exit and freedom, rather than the certain doom of a pit or a ghost:

The one sign of what lies behind the corridors is the wind: a pit produces a strong breeze (S) and an exit produces a weak breeze (W), while a ghost doesn't produce any breeze at all. Unfortunately, Pacman cannot measure the strength of the breeze at a specific corridor. Instead, he can stand between two adjacent corridors and feel the max of the two breezes. For example, if he stands between a pit and an exit, he will sense a strong (S) breeze, while if he stands between an exit and a ghost, he will sense a weak (W) breeze. The measurements for all intersections are shown in the figure below.

Also, while the total number of exits might be zero, one, or more, Pacman knows that two neighboring squares will not both be exits.



Pacman models this problem using variables X_i for each corridor i and domains P, G, and E

(a) [3] State the binary and/or unary constraints for this CSP (either implicitly or explicitly).

(b) [3] Suppose we assign X1 to E. Perform forward checking after this assignment. Also, enforce unary constraints.

X_1			E
X_2	P	G	E
X_3	P	G	E
X_4	P	G	E
X_5	P	G	E
X_6	P	G	E

(c) [3] Suppose forward checking returns the following set of possible assignments:

X_1	P		
X_2		G	E
X_3		G	E
X_4		G	E
X_5	P		
X_6	P	G	E

According to MRV, which variable or variables could the solver assign first?

(d) [3] Assume that Pacman knows that X_6 = G. List all the solutions of this CSP or none if no solutions exist.

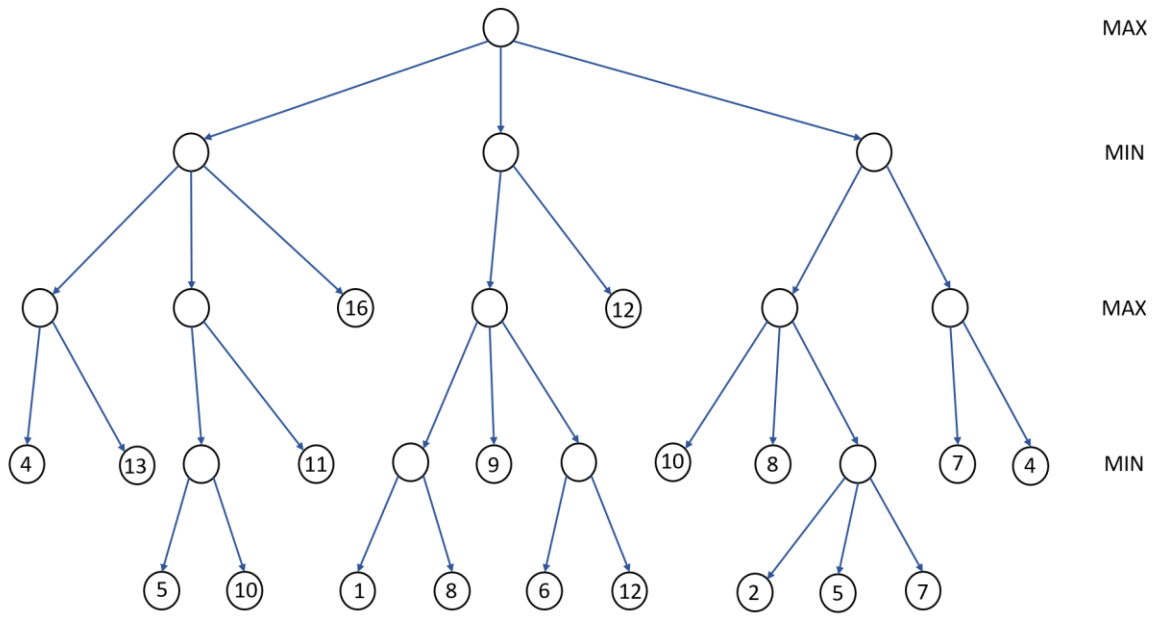
X_1	P		
X_2		G	E
X_3		G	E
X_4		G	E
X_5	P		
X_6	P	G	E

According to MRV, which variable or variables could the solver assign first?

(e) [3] Assume that Pacman knows that X_6 = G. List all the solutions of this CSP or none if no solutions exist.

Question 2: [15 Marks]

(a) [4] Use the Minimax algorithm to compute the minimax value at each node for the game tree below.



(b) [9] Use the Alpha-Beta pruning algorithm to prune the game tree above assuming child nodes are visited from left to right. Show all final alpha and beta values computed at the root, each internal node visited, and at the top of pruned branches. Note: Follow the algorithm in Figure 5.7 in the textbook. Also show the pruned branches.

(c) [2] For a general game tree (i.e., not limited to the above tree), are there any cases that the Alpha-Beta algorithm gives a different value at the root node than the Minimax algorithm? If yes, show an example; if no, just say no.

Question 3: [15 Marks]

Suppose three documents 1, 2 and 3 contain only five words "I", "am", "we", "are" and "groot". The following table summarizes the number of occurrences of each word in each document. Suppose one document is chosen at random (each document with equal probability), then one word in that document is chosen at random (each word with equal probability within that document).

Document / Word	*I*	*am*	*we*	*are*	*groot*
1	12	12	1	1	13
2	17	17	0	0	17
3	14	14	2	2	15

Note that the table can be used to compute conditional probabilities. For example, $P(Word = groot \mid Document = 1) = \frac{13}{12 + 12 + 1 + 1 + 13}$

(a) [2] What is the probability that the word is "*groot*", i.e., $P(Word = groot)$?
(b) [2] Suppose the randomly chosen word is "*we*". What is the probability that it comes from document 1, i.e., $P(Document = 1 \mid Word = we)$?
(c) [2] Suppose the randomly chosen word *starts with* an "*a*". What is the probability that it comes from document 2, i.e., $P(Document = 2 \mid Word = am$ or $Word = are)$?

Now suppose document 1 is chosen with probability P(Document = 1) = $\frac{1}{6}$, document 2 is chosen with probability P(Document = 2) = $\frac{1}{3}$, and document 3 is chosen with probability P(Document = 3) = $\frac{1}{2}$. Then, one word in that document is chosen at random (each word with equal probability within that document).

(d) [3] What is the probability that the word is "*groot*", i.e., $P(Word = groot)$?
(e) [3] Suppose the randomly chosen word is "*we*". What is the probability that it comes from document 1, i.e., $P(Document = 1 \mid Word = we)$?
(f) [3] Suppose the randomly chosen word *starts with* an "*a*". What is the probability that it comes from document 2, i.e., $P(Document = 2 \mid Word = am$ or $Word = are)$?

Show your calculations and round all final answers to 4 decimal places.

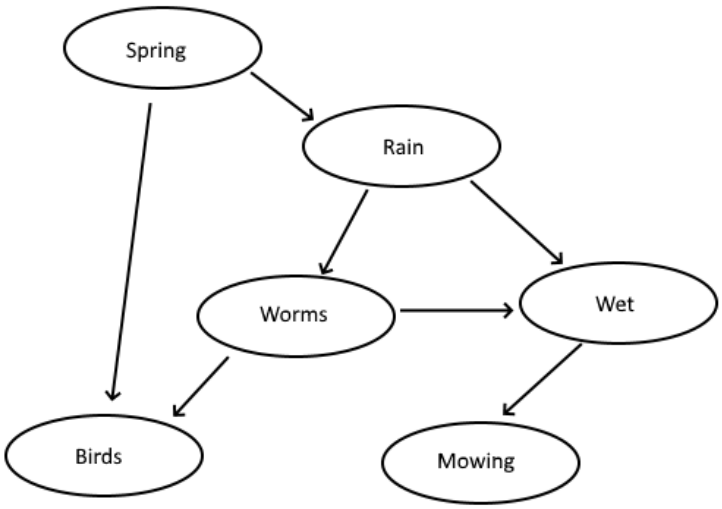
Question 4: [20 Marks]

Use the Bayesian Network below containing 6 Boolean random variables to answer the following questions using inference by enumeration. For the problems, Spring = S, Rain = R, Worms = Wo, Wet = We, Mowing = M, and Birds = B. Give your answers to 4 decimal places. Show your work.

(a) [4] What is the probability that it rains? ($P(R)$)
(b) [4] What is the probability that the grass is wet? ($P(We)$)
(c) [4] Given that it is spring, how likely is there to be someone mowing? ($P(M \mid S)$)
(d) [4] Given that it is spring, how likely is there to be a bird on the lawn? ($P(B \mid S)$)
(e) [4] If there are birds on the lawn, but there are no worms out, how likely is it that it is spring? ($P(S \mid B, \neg Wo)$)

The CPTs are defined as follows:

- $P(S) = 0.25$
- $P(R \mid S) = 0.7$, $P(R \mid \neg S) = 0.3$
- $P(Wo \mid R) = 0.7$, $P(Wo \mid \neg R) = 0.2$
- $P(We \mid R, Wo) = 0.12$, $P(We \mid R, \neg Wo) = 0.25$, $P(We \mid \neg R, Wo) = 0.1$, $P(We \mid \neg R, \neg Wo) = 0.08$
- $P(M \mid We) = 0.02$, $P(M \mid \neg We) = 0.42$
- $P(B \mid S, Wo) = 0.8$, $P(B \mid S, \neg Wo) = 0.4$, $P(B \mid \neg S, Wo) = 0.4$, $P(B \mid \neg S, \neg Wo) = 0.4$



Question 5: [15 Marks]

About 2/3 of your email is spam so you downloaded an open-source spam filter based on word occurrences that uses the Naive Bayes classifier. Assume you collected the following regular and spam mails to train the classifier, and only three words are informative for this classification, i.e., each email is represented as a 3-dimensional binary vector whose components indicate whether the respective word is contained in the email.

study	free	money	Category
1	0	0	Regular
0	0	1	Regular
1	0	0	Regular
1	1	0	Regular
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam

(a) [5] You find that the spam filter uses a prior p(spam) = 0.1. Explain (in one sentence) why this might be sensible and give the following model parameters when estimated as maximum-likelihood with add-one smoothing (i.e., using pseudo counts of one).

(b) [5] Based on the prior and conditional probabilities above, give the model probability P(spam|s) that the sentence s="money for psychology study" is spam

(c) [5] What should be the value of the prior p(spam) if we would like the above sentence to have the same probability as being spam as not spam, i.e., it would be classified as spam with probability 0.5?

Question 6: [10 Marks]

Ecology explores relationships between organisms and other living (biotic) things or non-living (abiotic) components in their environment. Certain factors may impact the abundance, distribution, or physiology of organisms, including abiotic factors such as temperature, moisture, or sunlight, and biotic factors such as the presence of predators or competitors. Linear regressions can be used to quantify these relationships and predict organism responses to various levels of the abiotic or biotic factor. Ecologists tested the grazing pressure of green crabs on clams. They constructed $1m^2$cages and planted 300 clams in each cage. Two days later, they counted the number of remaining clams and recorded the data in the data below.

Number of Crabs	Number of Clams Remaining
2	137
4	70
2	184
5	0
4	35
0	297
3	122
5	1
1	253
3	150

First, perform a linear regression using the step-by-step instructions for calculating m (slope) and b (intercept) of the regression line. **What is the full equation for the regression line that you calculated?**

Question 7: [10 Marks]

A hospital wants to predict the likelihood of patients developing complications after a surgical procedure based on various factors such as age, pre-existing health conditions, and type of surgery. They have collected data on 500 patients who underwent the same surgical procedure in the past year. For each patient, they recorded their age, presence of diabetes, presence of hypertension, whether they smoke, and the type of surgery performed (coded as 1 for minor surgery and 2 for major surgery), as well as whether they experienced complications post-surgery (coded as 1 for yes and 0 for no).

Age	Diabetes	Hypertension	Smoker	Surgery Type	Complications
45	0	1	1	1	0
60	1	1	0	2	1
55	1	0	1	1	1
70	0	1	0	2	1
50	1	1	1	1	0

use the model to predict the probability of complications for a new patient who is 60 years old, has diabetes and hypertension, is a smoker, and is undergoing major surgery.